

SpaceX Falcon 9 First-Stage Landing Prediction

IBM Data Science Capstone Project

Final Project Report | Dataset period: 2010-2020

Executive Summary

This project evaluates whether a SpaceX Falcon 9 first stage will land successfully. Recovery can reduce launch cost, so prediction supports competitive bid estimates. The work combines API collection, web scraping, wrangling, EDA, SQL, Folium, Plotly Dash, and supervised machine learning.

Key measure	Result
Falcon 9 launches	90
Prepared-label success share	87.8%
Most active site	CCAFS SLC 40 - 55
Largest share of successful launches	KSC LC-39A - 41.7%
Model test accuracy	83.3%

Main Findings

- Landing success increases over the observed period.
- Site and orbit are associated with different success patterns.
- Decision tree led cross-validation, but all four models scored 83.3% on the 18-launch test set.

1. Introduction and Business Problem

SpaceX advertises lower launch prices partly because Falcon 9 first stages can be recovered and reused. A competitor or customer can improve bid estimates by predicting whether recovery is likely. The question is: which mission characteristics are associated with successful landing, and how accurately can success be predicted?

Objectives

- Collect records from the SpaceX API and Wikipedia.
- Create a binary landing-success label.
- Explore site, orbit, payload, time, and success.
- Build geographic and dashboard views.
- Compare four classification algorithms.

Workflow

API and web scraping -> wrangling -> EDA and SQL -> Folium and Dash -> feature engineering -> model tuning and evaluation.

2. Data Collection and Preparation

The API workflow produced 90 Falcon 9 launches with 17 analytical fields. BeautifulSoup extracted 121 historical launch rows from Wikipedia. Missing payload masses were replaced by the mean; LandingPad nulls were retained because they represent no pad use. Landing outcomes were mapped to a binary Class label.

Launch site	Launches
CCAFS SLC 40	55
KSC LC 39A	22
VAFB SLC 4E	13

3. Exploratory Data Analysis

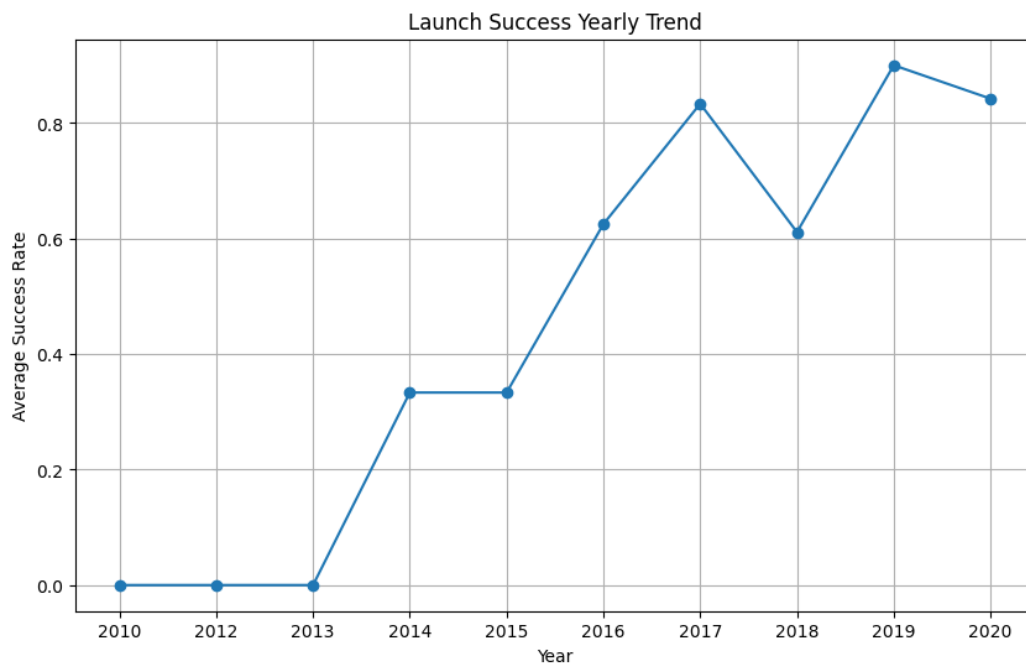


Figure 1. Falcon 9 launch success trend by year.

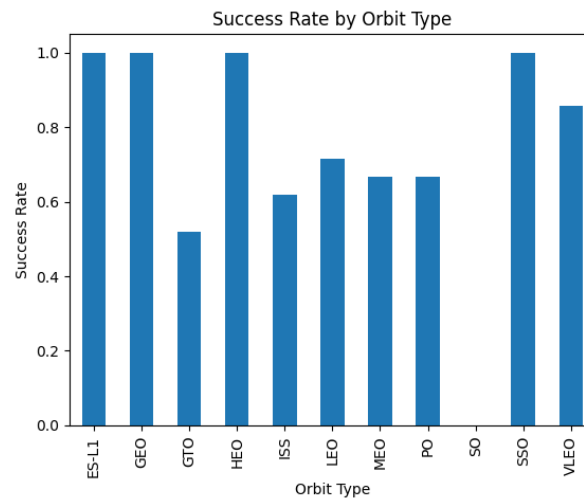


Figure 2. Success rate by orbit type.

Success improves as the program matures, while orbit types show distinct outcome profiles. The most common orbits are GTO (27), ISS (21), VLEO (14), and PO (9).

4. SQL Analysis

Question	Result
NASA CRS payload mass	45,596 kg
Average F9 v1.1 payload	2,928.4 kg
First ground-pad success	22 Dec 2015
Drone-ship success with 4,000-6,000 kg payload	B1022, B1026, B1021.2, B1031.2
Drone-ship successes, 2010-2017	5

SQL establishes operational milestones and isolates mission groups by payload, booster, outcome, and date.

5. Folium Geospatial Analysis

The map marks launch sites, success and failure clusters, and distances to coastline, city, railway, and highway points. Sites are near coastlines and transportation infrastructure.

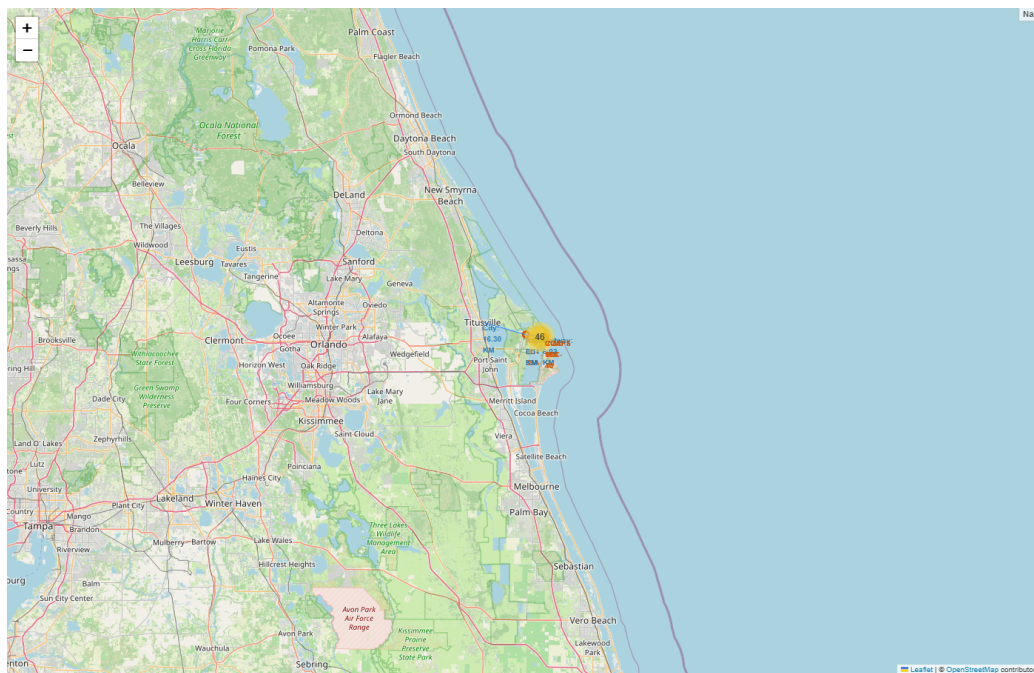


Figure 3. Florida launch-site clusters and proximities.

6. Plotly Dash Analysis

The dashboard supports site selection, success pie charts, a payload range slider, and outcome scatter plots colored by booster category.

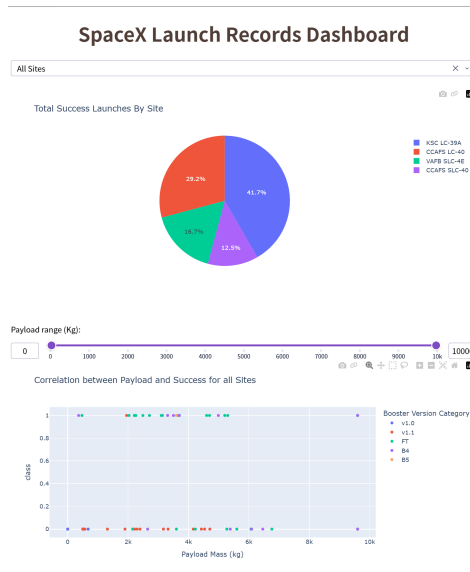


Figure 4. All sites: KSC LC-39A accounts for 41.7% of successful launches.

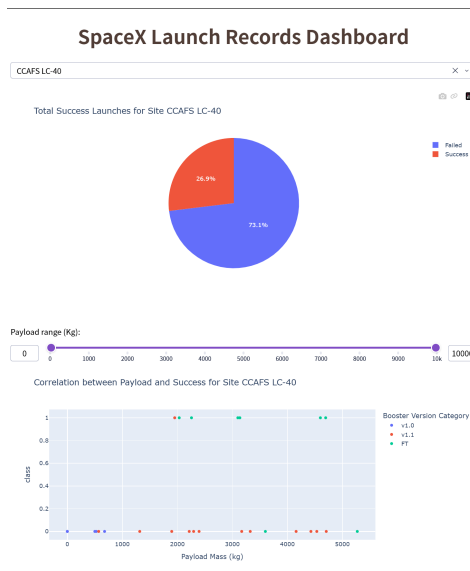


Figure 5. CCAFS LC-40: 26.9% success and 73.1% failure.

7. Predictive Analysis

One-hot encoded features were standardized and split 80/20 with random_state=2. GridSearchCV with 10 folds tuned four classifiers.

Model	CV accuracy	Test accuracy
Logistic regression	84.6%	83.3%
Support vector machine	84.8%	83.3%
Decision tree	92.9%	83.3%
K-nearest neighbors	84.8%	83.3%

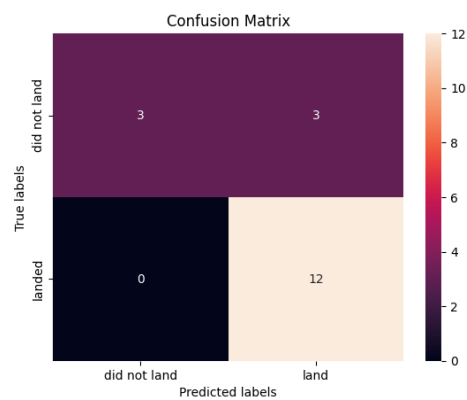


Figure 6. Decision-tree confusion matrix.

Decision tree led cross-validation, but identical test accuracy means the 18-case test set is too small to establish a definitive winner.

8. Conclusions and Recommendations

- Landing performance improved substantially over time.
- Site, orbit, payload, booster block, reuse count, grid fins, and legs provide useful signal.
- KSC LC-39A leads the successful-launch distribution; CCAFS LC-40 shows 26.9% success in the site view.
- All models achieved 83.3% test accuracy; decision tree led cross-validation at 92.9%.

Recommendations

- Expand the dataset with newer launches.
- Use repeated or nested cross-validation.
- Evaluate calibrated probabilities and cost-sensitive errors.
- Retain the dashboard and map for stakeholder scenario exploration.

GitHub Repository

<https://github.com/TristinWen/spacex-falcon9-landing-prediction>

The public repository contains the completed notebooks, Dash source, screenshots, map, presentation, and report. All results come from the user-provided executed files under C:/Users/Administrator/Downloads/completed.